

Following the Speedup: An Audit of Adaptive Draft Length in Speculative Decoding

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Abstract—Adaptive draft length promises a double-digit speedup, and the published strategies do not deliver it. We establish a tuned fixed-length baseline, then a Bayes ceiling on what any draft-side signal could add on top of it, and most of the promised gain turns out to sit above that ceiling: unreachable, not merely unreached. The practical consequence is blunt. Reported adaptive gains fall below the ceiling, and a tuned fixed draft length already reaches that far, so the effort spent on signal-based adaptive length buys little that tuning does not. We show this by following the promised gain through three gates. Published oracle arguments put the prize at +12 to 23% over any fixed draft length. Gate one asks whether the gain exists: a Bayes-ceiling decomposition attributes about 80% of it to verification-outcome variance that the draft-side state does not determine, which leaves roughly +3% for any policy reading that state. Gate two asks what could reach the remainder. Cheap logit features recover almost none of it and token position recovers exactly none, but the drafter’s hidden state does carry the signal, at AUC 0.84 to 0.88. Gate three puts that signal in a running engine, where acting on it costs 2 to 17% against a tuned fixed depth. One policy survives the audit: signal-free saturation tail-pruning, worth +2 to 6% at batch size 1, with one of three workloads a wash and the gain gone by batch 8. Seeing the leak at all required fixing six evaluation practices that inflate adaptive-length results, including unpaired GPU timing that gave us three contradictory answers from identical configurations. We release the protocol, the code, and the raw per-run measurements.

Index Terms—speculative decoding, adaptive draft length, per-step acceptance signal, hidden-state probe, oracle decomposition, benchmarking methodology, EAGLE, efficient inference, LLM serving.

I. INTRODUCTION

Speculative decoding (SD) speeds up generation by having a cheap draft model propose K tokens that the target then verifies in a single parallel pass [1], [2]. The draft length K is the primary tuning knob. Too short a draft leaves speedup unclaimed; too long a draft spends compute on tokens the target will reject. A large literature therefore proposes setting K per step from some decode-time signal.

The case for doing so rests on an oracle argument. Suppose the number of tokens the target will accept were known in advance, and the drafter proposed exactly that many. Throughput would rise by double digits over any fixed K . SpecDecode-Bench [18] recently made this concrete on vLLM, establishing that verification dominates SD execution, that accepted length swings widely across positions and requests, and that a wide

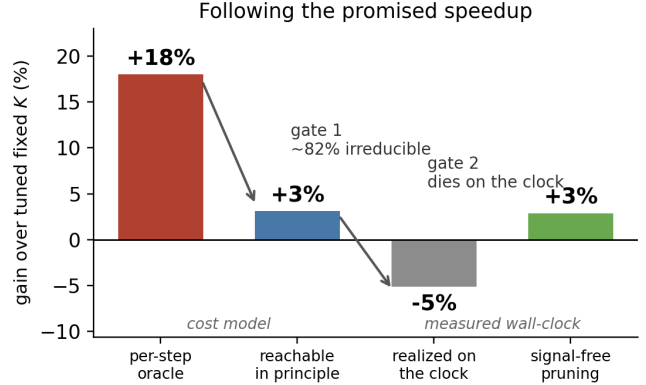


Fig. 1. The audit in one picture. The oracle promises +18% on this model. Most of it is irreducible from any draft-side signal, and what remains inverts sign once a real engine has to execute the policy. The survivor at right reaches a similar height by a different route, using no learned signal at all. The first two bars are cost-model figures on the Llama-3.1-8B ladder and the last two are measured wall-clock, which is the point: the promise is scored in one currency and the delivery in another.

gap separates deployed methods from the oracle. It closed with a call for “a lightweight predictor that can adaptively select...approaching the oracle bound.”

We began by trying to build a better draft-length policy, following where the research was heading. What stopped us was more basic. We could not reproduce the speedups the papers reported. In our own harness, Python-side overhead was large enough to swamp the effect we were trying to measure. The numbers we could produce matched neither the published ones nor what vLLM already delivers. So the question changed. Before asking which policy wins, we had to work out how to measure speedup honestly, how much is genuinely available, and why the field is moving in the direction it is. This paper is that measurement, and Fig. 1 is its whole argument in one picture: the promised gain, and what each stage of the audit removes from it.

The oracle number is the speedup available to a drafter that knows exactly how many of its proposed tokens the target will accept. That is knowledge of the future, not a policy, and the distinction carries most of what follows.

A. The audit

We follow the claimed number through three gates, each of which removes part of it (Fig. 1).

Gate 1, is the gain there at all? We decompose the per-step oracle into what a tuned fixed K already collects, what a draft-side signal could still reach, and what is irreducible. About 80% falls in the last bucket. The oracle overstates the achievable prize by roughly five times (§V).

Gate 2, can any real signal reach the rest? The reachable slice is genuinely there, and we find where it lives. It is not in the cheap logit features the literature has used, which recover -2.7 to 8.5% of the span and go negative on the reasoning head, nor in token position, which recovers exactly zero. It is in the drafter’s hidden state, from which a PCA and logistic probe recovers about a fifth of the oracle span (§VI). This closes a door rather than opening one. We looked in the best remaining place and the answer was still no.

Gate 3, does it survive execution? We put the signal in a live engine. Every learned or entropy-based stopping policy we tested loses to a tuned fixed depth, by 2 to 17% (§VII). The offline arithmetic and the wall clock disagree because drafting is cheap and verification is not, so stopping early saves little and forfeits a lot.

What survives is one policy that uses no signal: stop drafting when the cumulative path probability saturates. It is worth $+2$ to 6% at batch size 1, and the mechanism explains why it works where prediction fails (§VIII).

B. Why this was not visible earlier

The effects in this area are a few percent, and at that scale the measurement apparatus decides the answer. Six practices, each common and each individually reasonable, inflate adaptive-length results. We encountered all six. The worst was unpaired sequential timing, which returned $+2.4\%$, -1.9% and $+6.4\%$ from identical configurations before we started pairing. Table I lists them with what each one did to our own numbers. The protocol we ended up with is the instrument that made the rest of the audit readable, so we describe it before the results (§IV) rather than as a separate finding.

Every comparison here concerns speed at unchanged output, since exact-verification SD is lossless.

II. RELATED WORK

Feature-based drafting. EAGLE-1/2/3 [3]–[5] draft from the target’s hidden states. EAGLE-2 and 3 adapt the tree shape using draft confidence but still commit to a static depth. TALON [15] fits tree structure to a token budget.

Signal-based length control. SVIP [6], AdaEDL [10] and DSDE [11] stop drafting on entropy, margin or KL thresholds.

Learned controllers. SpecDec++ [7] trains an acceptance head and DISCO [8] a classifier. BanditSpec [9] and its successors treat K as a bandit arm.

Cumulative-confidence stopping. SADDLE [13], PACER [14] and DDD [12] stop on accumulated acceptance probability. This is the class our audit leaves standing.

Serving-level adaptation. Nightjar [16] and TurboSpec [17] gate speculation by system load. That is a different lever, and a profitable one.

Benchmarking. SpecDecode-Bench [18] established verification dominance, the oracle gap and acceptance variance on vLLM. We credit those findings and claim only our delta: splitting that gap into reducible and irreducible parts, locating the reducible part, testing it on a clock, and the protocol that made the test trustworthy.

Baselines. Most adaptive- K papers compare against $K=1$ or an untuned default [6], [7], [10], [12]. Measured against a per-workload tuned fixed K , their reported 2 to 8% gains fall inside the $+2\%$ per-request oracle we measure (§IX).

III. SETUP

Models. Llama-3.1-8B-Instruct with yuhuili/EAGLE3-LLaMA3.1-Instruct-8B; Qwen3-14B with AngelSlim/Qwen3-14B_eagle3; DeepSeek-R1-Distill-Llama-8B with yuhuili/EAGLE3-DeepSeek-R1-Distill-LLaMA-8B.

Engine. vLLM 0.23 (V1) on an H100, greedy, $B=1$ unless stated. Verification is exact, so decoding is lossless; fp32 outputs are token-identical to the non-speculative baseline. The wall-clock experiments use the SafeAILab/EAGLE-3 reference implementation because vLLM exposes no per-step draft-length hook.

Workloads. HumanEval, GSM8K and MT-Bench. Sample sizes differ by experiment and we state them explicitly, since they are small. The hidden-state capture behind the decomposition uses 30 prompts per workload at 128 max new tokens, which yields 41K to 60K labelled draft positions per model. The wall-clock experiments use 10 prompts per workload for threshold selection and probe training and a disjoint 10 for benching, also at 128 max new tokens, with 8 per workload in the batched runs. Repetition rather than prompt count carries the precision in the timing results: 3 repeats in the policy zoo, 4 rotated cycles under the paired protocol, and 3 fresh containers for the fair-baseline replication. Competition math (MATH-500/AIME) serves as a reasoning control, and QuALITY 8K passages as a long-context one.

Baseline. The best fixed K by grid search over $K \in \{1..7\}$, tuned per workload. On Llama-3.1-8B mixed traffic this lands at $K=2$ and $1.54\times$ measured net speedup. This is a deliberately hard baseline, and most of what follows depends on it being hard.

There is a fair objection to it, which we should meet before the results rather than after. Tuning K per workload assumes the workload is known in advance, and a deployment serving mixed traffic does not know that. On this reading our baseline gets something for free that an adaptive method has to earn, and the comparison is unfair to adaptation. Our own data answers this. The per-request oracle, which is exactly the gain available to a method that identifies the right depth for each request without being told, is $+2.0\%$ over the globally tuned K . Even perfect per-request adaptation therefore recovers about two percent, so a tuned fixed depth is not merely a convenient baseline; it is close to the best that request-level knowledge can buy. Where per-workload tuning is genuinely

impossible, that +2.0% is the prize on offer, and §IX shows online bandits failing to collect it.

Capture. Hooks on the EAGLE3 draft head’s `compute_logits` record the full per-position draft hidden state, 4096 or 5120 dimensions, 41K to 60K positions per model, with per-step acceptance labels $\text{accept}[j] = [\text{acc} > j]$.

Two oracles. We use two, and they differ by a lot, so we separate them here. The *per-step* oracle may change K at every decode step, knowing exactly how many tokens the target will accept at that step. It is worth +12 to 23% over a tuned fixed K and it is the number the literature’s motivating plots show. The *per-request* oracle must commit to a single K for a whole request, but gets to pick the best one for that request. It is worth only +2.0%. The gap between the two is the value of reacting within a request, and §V is about how much of that value any real policy can capture.

Notation. K is the draft length in the chain regime, or tree depth. For a decode step, $\text{acc} \in \{0, \dots, K\}$ counts the drafted tokens the target accepts, so the step emits $\text{acc}+1$ tokens, and MAT is $\mathbb{E}[\text{acc}+1]$. The constant C is the draft-to-target per-token cost ratio: one draft forward costs C target forwards, so a K -token step costs $1+CK$.

Cost model. Offline policies are scored with $\text{speedup} = \text{MAT}/(1+CK)$. We fit C instead of assuming it. Setting $\text{net}(K) = \text{accept_len}(K)/(1+CK)$ against the measured fixed- K throughput sweep reproduces H100 throughput to about 1% at $C \approx 0.066$ to 0.072 . Headline offline numbers are reported at both $C=0.15$, a conservative proxy, and the fitted $C=0.072$. The conclusions do not move.

One property of this formula matters for everything that follows: it charges nothing for deciding when to stop. A fixed draft length pays no such cost, while any signal-based policy has to compute its signal, test it and branch on it at every step. Every offline adaptive number in this paper is therefore an upper bound on what that policy could achieve in a real engine, and the gap between the two is the subject of §VII.

IV. HOW WE MEASURED

This section is the instrument, not a result. We describe it first because every number after it depends on the apparatus, and because the apparatus was the hardest part of the work to get right.

Our first attempts produced unstable answers, and in one case directly contradictory ones. Diagnosing that instability surfaced six distinct ways an adaptive-length evaluation can return a publishable-looking number that does not survive contact with a fair test. Table I lists them alongside what each one did to our own measurements.

Three of the six deserve a note here. Selecting a stop threshold on the same prompts used for benchmarking inflates results substantially, and the inflation is not small: the same policy reads +6.4% in-distribution and −1.9% held-out. Splitting those prompts contiguously brings its own problem on category-ordered suites like MT-Bench, where the best fixed depth itself flips from 7 to 5 across the split, tangling

distribution shift with policy generalisation. We use interleaved splits.

The third took the most effort to diagnose. Benchmarking arms in sequential blocks is unreliable at the effect size in question. Within-container throughput drifts by several percent between blocks, and error bars taken from consecutive repeats cannot see that drift because they measure the wrong thing. Three runs with identical thresholds gave us +2.4%, −1.9% and +6.4%. Our final protocol benches all configurations round-robin, rotates the order each cycle, and takes per-cycle paired differences, which cancels drift by construction (Fig. 3). Statistics throughout are paired per-cycle differences against the strongest baseline arm, reported as mean $\pm$ standard error over cycles. Where we say “strongest,” we mean the arm that minimises our own reported gain.

V. GATE 1: IS THE HEADROOM THERE?

Start where the literature starts, with the oracle number. It is computed by looking at what the target actually accepted and drafting exactly that far, which makes it a measure of hindsight rather than of any policy. The question this gate asks is how much of that hindsight a predictor could ever have had in advance.

We answer it by splitting the oracle gap into three parts: what a tuned fixed K already collects, what a draft-side signal could still reach, and what is left over. The last part is the interesting one, because nothing on the draft side can touch it. To measure the split we build a ladder of five rungs, each one a policy with strictly more information than the last. All five run under 8-fold cross-validation split by generation, with thresholds chosen on train:

- **best fixed K** , the tuned static baseline;
- **Bayes(position)**, the best stop-policy given only the position-survival curve. Position is a static signal, so this rung must collapse to a fixed K . It is a built-in correctness check;
- **probe (deployable)**, a stop-policy on an out-of-fold PCA-50 and logistic probe over the full hidden state, with the threshold chosen on training generations;
- **probe (oracle threshold)**, the same probe scores on the same held-out rows, with the threshold chosen to maximise performance on those rows. It removes threshold-selection error only, so it bounds this probe family and not every possible reader of the hidden state. Since it differs from the rung above by the threshold alone, it must bound it in every fold, which is a second built-in check;
- **per-step oracle**, $K = \text{acc} + 1$, unreachable by construction.

Table II and Fig. 2 give three readings.

Bayes(position) is zero everywhere. A static signal cannot beat a tuned fixed K , and the fact that this rung collapses exactly validates the pipeline. It also explains a puzzle in the literature: position and entropy carry AUC around 0.5 to 0.58, which bought earlier methods nothing because there was nothing there to buy.

TABLE I

SIX WAYS TO MEASURE AN ADAPTIVE-LENGTH POLICY AND GET THE WRONG ANSWER. EVERY ROW IS A CASE WHERE THE DEFECTIVE PROTOCOL RETURNED A PLAUSIBLE NUMBER; THE LAST COLUMN IS WHAT THE CORRECTED PROTOCOL DID TO IT. AT THE $\pm 5\%$ SCALE THESE CLAIMS OCCUPY, NONE OF THE SIX IS OPTIONAL.

#	Failure mode	How it shows up	What it did to our number
1	Mechanism-mismatched simulation	Offline chain model scores a tree engine	Chain +2 to 3% <i>inverts sign</i> on deployment (§VII)
2	In-distribution threshold selection	Threshold tuned on the benched prompts	Tail-pruning +6.4% $\rightarrow$ -1.9% held-out (Table IV)
3	Unpaired sequential benchmarking	Arms timed in blocks under container drift	Identical configs gave +2.4%, -1.9%, +6.4%
4	Intra-run throughput step-changes	Clock shifts <i>within</i> a cycle; pairing cannot cancel it	Fixed arms jumped 14 to 18% mid-cycle; use ≥ 6 cycles
5	Dead-code instrumentation	Patching a shipped but unused codepath	Silent no-op; first vLLM result quarantined
6	Padded-budget baselines	Baseline verifies slots it would not verify natively	Motivated the 3-engine rerun; verify flat, <1%
<i>Split ordering, a rider on #2</i>		Contiguous split is non-iid on category-ordered suites	Best fixed depth itself flipped 7 $\rightarrow$ 5 on MT-Bench

TABLE II

THE DECOMPOSITION. ROWS 1–2 ARE INSTRUCTION MODELS, ROW 3 A REASONING MODEL; ROWS 4–5 ARE THE SAME INSTRUCT AND REASONING PAIR ON COMPETITION-MATH CoT. **UNITS DIFFER BY COLUMN.** ORACLE SPAN AND NET ARE ABSOLUTE COST-MODEL GAINS OVER THE TUNED FIXED K ; THE TWO RECOVERY COLUMNS ARE FRACTIONS OF THAT SPAN, SO A RECOVERY LARGER THAN A SPAN IS NOT A VIOLATION. NET IS THE DEPLOYABLE PROBE’S OWN GAIN OVER THE TUNED FIXED K , COMPUTED PER FOLD AND AVERAGED. ORACLE SPAN IS COMPUTED PER FOLD RATHER THAN ONCE OVER THE WHOLE CAPTURE; THE WHOLE-CAPTURE CONVENTION SHIFTS THE RECOVERY COLUMNS BY AT MOST 1.6 PP AND CHANGES NO CONCLUSION. BAYES(POSITION) IS OMITTED BECAUSE IT IS $+0.0\%$ IN ALL FIVE SETTINGS BY CONSTRUCTION, WHICH IS THE CORRECTNESS CHECK DESCRIBED ABOVE.

Model	Oracle span	Recovery (% of span)		Net (abs.)
		probe	oracle thr.	
Llama-3.1-8B	+18.1	16.8 ± 2.7	18.2 ± 2.6	$+3.1 \pm 0.6$
Qwen3-14B	+12.4	17.2 ± 2.4	19.5 ± 2.4	$+2.0 \pm 0.3$
DeepSeek-R1	+4.5	-0.3 ± 0.2	0.7 ± 0.5	-0.0 ± 0.0
Llama-3.1-8B, math	+23.2	9.5 ± 3.7	12.0 ± 3.8	$+2.5 \pm 1.0$
DeepSeek-R1, math	+8.5	-1.3 ± 0.7	1.9 ± 0.6	-0.1 ± 0.1

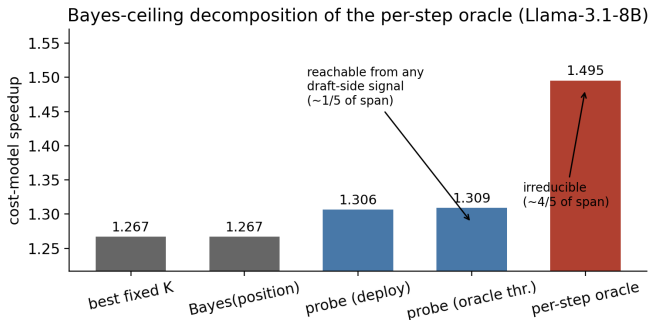


Fig. 2. The five rungs for Llama-3.1-8B. Only about a fifth of the oracle span is reachable from any draft-side signal, and Bayes(position) lands exactly on the tuned fixed K , which is the correctness check the ladder has built into it.

The probe with an oracle threshold sits close to the deployable probe and far below the oracle. Even handed a perfect threshold, a hidden-state probe recovers only about a fifth of the span. Perfect threshold choice would add between 1.0 and 3.2% of the span, and we should be careful with that number, since the deployable and oracle-threshold intervals overlap in every setting. Treat it as a direction, not a settled result. The

other four fifths is verification-outcome variance that no draft-side signal can see. The oracle overstates what is achievable by roughly five times.

The reachable part is real for the instruction models, worth +2.0 to 3.1% net over a tuned fixed K , and absent for the reasoning head.

A. Controls

Four checks back the decomposition. *Permutation*: shuffling accept labels across draft blocks, preserving marginals and prefix structure, collapses Llama recovery from $+16.8 \pm 2.7\%$ to $-5.6 \pm 1.4\%$, a 22.4pp drop. Both arms use the same paired-within-fold estimator, so they differ only in whether labels belong to their hidden states. That is signal, not leakage. *Independent re-derivation*: fresh code with different folds, PCA dimension, seed and policy reproduces the oracle, fixed and ceiling rungs to under 0.1pp and reproduces the sign of every recovery. Magnitude is implementation-sensitive. *Measured-cost grounding*: re-scoring at the fitted $C=0.072$ leaves Llama at +17.3% recovery and +2.9% net. *Detection*: the probe we report reads AUC 0.84 to 0.88 across the three models, and the weaker probe families of §VI run down to 0.79, against 0.484 for an earlier 16-dimensional random projection. The weak probe was the earlier limitation, not the hidden state.

B. A confound we could not remove

On identical competition-math inputs, Llama-3.1-8B keeps its signal at $+9.5 \pm 3.7\%$ recovery and $+2.5 \pm 1.0\%$ net, while DeepSeek-R1 shows none. That looks like a boundary between model classes, and we cannot claim it is. The DeepSeek head is simply weaker: accept-length 1.47 against 2.40 on the same task, position-0 acceptance 0.28 against 0.55. Detection is not what fails here: the probe reads this head at AUC 0.87, as well as it reads either instruction head. What a weak head yields is a small ceiling, and a fifth of a +4.5% span is nothing worth collecting. That is the same detection-versus-exploitation gap the rest of the audit turns on, and head strength produces it whatever the model is doing. We therefore scope the claim to this head. Separating “reasoning model” from “weak draft head” needs a strong reasoning-model EAGLE head, and none exists today.

VI. GATE 2: WHERE THE SIGNAL LIVES

Gate 1 leaves about a fifth of the span on the table. Gate 2 asks what can reach it, and the answer narrows quickly.

Cheap logit features are the natural first guess, and they are where most published methods look. Entropy, margin and SVIP-style statistics recover between -2.7 and 8.5% of the span across the same three models, and the reasoning head is the negative one. Token position recovers exactly nothing, as Gate 1’s correctness check already implied. The drafter’s hidden state is different. A PCA-50 and logistic probe over the full state recovers $+16.8 \pm 2.7\%$ and $+17.2 \pm 2.4\%$ of the oracle for the two instruction models, detecting rejection at AUC 0.84 and 0.88.

The probe family is not the limitation, and we checked it instead of assuming it. On the same captured traces we compared four probes. These were the 16-dimensional random projection with gradient boosting we started from, logistic regression on the full hidden state, a two-layer MLP (256 then 64 units) on the full hidden state, and PCA-50 with logistic regression. The random projection detects almost nothing, at AUC 0.484. The other three land between AUC 0.79 and 0.88, and the MLP does not beat PCA-50 with logistic regression on recovery, which is the probe we report throughout. Capacity was not the missing ingredient.

The scope of this result warrants care. It is a negative result with a precise address. We looked in the most informative place available on the draft side, using a probe strong enough to find the signal that weaker probes had missed, and the ceiling barely moved. Gate 2 therefore closes the search rather than opening a direction: if the hidden state yields only a fifth of the span offline, no cheaper signal will do better.

VII. GATE 3: DOES IT SURVIVE EXECUTION?

Offline arithmetic says a few percent is available. We now put it on a clock. vLLM 0.23 has no per-step draft-length hook, so these tests run in the SafeAILab/EAGLE-3 reference implementation, in eager PyTorch, with no CUDA-graph penalty. Fixed and adaptive arms share the same instrumented codepath, which equalises per-step overhead. Llama-3.1-8B, H100, greedy, otherwise as §III.

A. A controller on the step’s seed state

A PCA and ridge probe on the step’s seed hidden state sets the draft depth. Our first capture here was incorrect, and the nature of the error is worth recording: seeds were mispaired with accept-lengths across prompts, because one dangling unverified seed per prompt shifted the global index. A code audit caught it, the mismatch equalling the prompt count exactly confirmed it, and we corrected the pairing and reran. With correct pairing the seed carries modest signal, correlation 0.26. The controller still loses, by 5.7%, 2.4% and 5.5% against the best fixed depth on HumanEval, GSM8K and MT-Bench.

TABLE III
IN-LOOP PROBE AGAINST FIXED FULL DEPTH (TOK/S).

Workload	Fixed (full depth)	In-loop adaptive	Gain
HumanEval	144.5	137.2	−5.1%
GSM8K	128.4	121.8	−5.1%
MT-Bench	133.4	125.3	−6.1%

B. Stopping inside the draft loop

The faithful test is to break the depth loop the moment a per-level probe predicts rejection. We built a verified copy of the EAGLE-3 draft routine that does exactly this, capping the shorter tree safely, and confirmed it reproduces coherent generation. The level probe carries real signal, AUC 0.796, close to the 0.84 we measured offline. Early stopping still loses 5 to 6% (Table III).

The cause is economic rather than statistical. The draft head is cheap, and it produces tokens faster than the target can verify them, which is consistent with SpecDecode-Bench’s 42 to 95% verification share. That sets the shape of the trade. The speedup on a step is however deep the drafter gets before the target stops agreeing with it, and stopping early truncates exactly that. The drafting already done still has to be paid for, but it now buys fewer accepted tokens, so it stops being an investment and becomes overhead on top of the verify pass. Optimising away draft compute is the wrong move when drafting is the cheap half. The offline chain analysis said +2 to 3%; deployed into the tree engine it inverts sign. That is failure mode 1 in Table I, and the literature’s evaluation protocol permits it.

C. Three published-style policies

We extended the test to three stopping policies through the same verified codepath: SVIP-style entropy, SADDLE/PACER-style cumulative path probability, and a SpecDec++-style trained hidden probe. A margin-threshold arm ran in an earlier pass and was dropped before the held-out verification, so we do not report a number for it.

These are reimplementations of each stop *mechanism* rather than ports of the published systems, and the distinction matters for what the results can be read to mean. Running four released codebases against each other would confound the stop rule with every other difference between them: tree construction, draft-head training, engine, tokenizer handling. We instead hold all of that fixed and change one line, so the arms differ only in what triggers the stop. That buys comparability at the cost of fidelity. A published system may well outperform our version of its rule, and a loss in Table IV is evidence about the mechanism, not about anyone’s implementation of it.

Table IV gives the verdict, and in the same numbers shows why the protocol of §IV was necessary. Read in-distribution, three arms look like wins. Read on held-out prompts, every one of them loses.

That is the gate-3 verdict. Every learned or entropy-based per-step policy we tested loses to a tuned fixed depth, across

TABLE IV

THE POLICY ZOO (EAGLE-3 TREE, LLAMA-3.1-8B, $B=1$), AS % AGAINST THE STRONGEST FIXED DEPTH IN THE SAME RUN, MEAN OVER 3 REPEATS. PER-CELL SE RUNS 0.4 TO 1.7 PP. EVERY CELL CLEARS ITS OWN ERROR BAR EXCEPT THE TWO NOMINAL TIES, SATURATION TAIL-PRUNING ON HUMANEval HELD-OUT ($+0.1 \pm 0.9$) AND ENTROPY ON MT-BENCH IN-DISTRIBUTION ($+1.1 \pm 1.7$). THE TWO HALVES ARE THE SAME POLICIES, THE SAME HARDWARE AND THE SAME CODE. THEY DIFFER ONLY IN WHETHER THE STOP THRESHOLD WAS CHOSEN ON THE BENCHMARKED PROMPTS. UNDER IN-DISTRIBUTION SELECTION THE BEST FIXED DEPTH IS FLATTERED TOO, $K=8/6/5$ AGAINST $K=5/5/5$ HELD-OUT. THESE ARE UNPAIRED 3-REPEAT RUNS; FOR THE SURVIVOR, THE DRIFT-CONTROLLED NUMBERS IN TABLE V SUPERSEDE THEM.

Policy	HEval	GSM8K	MT-B	mean
<i>Held-out threshold selection (our protocol)</i>				
Entropy (SVIP-style)	-2.0	-4.3	-7.7	-4.7
Saturation tail-pruning	+0.1	-2.5	-3.2	-1.9
Hidden probe (SpecDec++)	-14.0	-14.7	-22.3	-17.0
<i>In-distribution selection (the common protocol)</i>				
Entropy (SVIP-style)	+0.9	+2.6	+1.1	+1.5
Saturation tail-pruning	+4.5	+7.3	+7.3	+6.4
Hidden probe (SpecDec++)	-15.1	-12.8	-16.2	-14.7

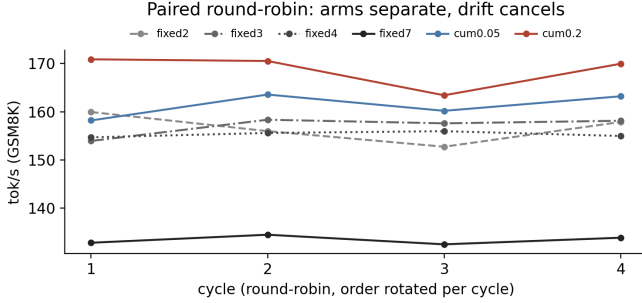


Fig. 3. Why pairing is necessary. Per-cycle raw throughput shows the arms separating and the container drifting at the same time; paired per-cycle differences cancel the drift that produced three contradictory sequential answers.

a range from 2 to 17%. The signal Gate 2 located is real and acting on it makes things worse.

VIII. WHAT IS LEFT STANDING

One arm in Table IV behaves differently from the rest, and it is the one that uses no learned signal at all. Saturation tail-pruning stops drafting when the cumulative best-path probability the drafter has already computed falls below a threshold. We write the rule as $\text{cum}(\tau)$: stop once that running product drops below τ , so a larger τ stops sooner and drafts less. Each engine gets its own τ chosen on train, $\text{cum}(0.05)$ in the EAGLE-3 tree and $\text{cum}(0.2)$ in the vLLM chain, and we name the value wherever we quote a number. Zero model cost, no training, and the policy family already exists in prior art [12]–[14]. Under the paired protocol it wins (Table V).

It wins at a realized mean depth of 5.62 against a fixed 7 to 8. Outputs are identical, the verified tree size is unchanged, and the fixed arms run the same instrumented codepath, so the gain is pure draft-iteration savings.

TABLE V

PAIRED VERDICT ON HELD-OUT PROMPTS AGAINST THE STRONGEST PER-WORKLOAD FIXED DEPTH. SATURATION TAIL-PRUNING, THRESHOLD 0.05 SELECTED ON TRAIN.

Workload	Paired gain	Cycles positive
GSM8K	$+3.58\% \pm 0.32$	4/4
MT-Bench	$+4.77\% \pm 0.50$	4/4
HumanEval	$+2.90\% \pm 1.83$	3/4

The decomposition explains why this one works when prediction does not, and the explanation comes down to what it costs to ask. Predicting acceptance requires anticipating re-alization variance, which §V showed is mostly irreducible, and the probe has to run PCA and a logistic model over the hidden state at every level to attempt it. Saturation pruning multiplies the current token probability into the running product for its branch, a quantity the tree search has already computed. Its decision is free.

That difference is measurable, not rhetorical. In the held-out zoo run the probe drafts to a realized depth of 6.95, which lets us compare it against a fixed depth of 7 doing the same drafting and the same verification. The probe is 10 to 13% slower on the three workloads (120.9 against 138.4 tok/s on HumanEval, 115.6 against 129.2 on GSM8K, 85.0 against 96.0 on MT-Bench). That gap is the cost of asking the question. Set against a reachable gain that Gate 1 caps at a few percent, the decision costs several times what the decision is worth. Saturation pruning wins the same comparison by not paying it.

A. How far the result travels

Cross-head. On the weaker DeepSeek-R1-Distill head, accept-length about 1.5 against Llama’s 2.4, the same protocol yields a tie: -0.7% , -0.9% and $+1.4\%$, none significant, at realized depth 6.19. It prunes about 0.8 levels where Llama prunes 1.4. Tail-pruning pays when a strong head makes deep fixed budgets frequently wasteful, and on a weak head there is less tail to prune. This run also surfaced failure mode 4: one cycle was contaminated by an intra-cycle throughput step-change, with the fixed arms jumping 14 to 18% mid-cycle, which pairing does not cancel. Robust use of the protocol wants six or more cycles, or median-of-cycles statistics.

Production engine. Since vLLM exposes no per-step hook, we patched the live draft loop of its V1 engine, locating it by stack-tracing from a hooked draft-head call. This was necessary because vLLM 0.23 ships a parallel legacy spec-decode package that is dead code in this configuration. Patching it produces a plausible, silent no-op that only an arm-separation check catches, and our first attempt was invalidated exactly this way and quarantined. All arms share the patched codepath and pay verification for all 7 slots, which isolates draft-side savings under equal verify cost. The gates hold: 19% separation between fixed arms, and the strongest fixed arm lands at $k=2$ to 3, matching the independently tuned production $K=2$. Against the strongest fixed arm, $\text{cum}(0.2)$

TABLE VI

FAIR-BASELINE REPLICATION: PAIRED GAIN OF CUM(0.2) OVER THE STRONGEST NATIVE ENGINE, EACH DRAFTING ITS TRUE K , ON THREE FRESH CONTAINERS. POSITIVE OR TIED IN 9/9 CELLS, AND THE WORST IS A TIE. EACH CELL IS THAT CONTAINER’S OWN PAIRED MEAN OVER 4 CYCLES; THE LAST COLUMN IS THE MEAN OVER THE THREE CONTAINERS WITH ITS STANDARD ERROR, THE SPREAD ACROSS CONTAINERS DIVIDED BY $\sqrt{3}$.

Workload	cont. 1	cont. 2	cont. 3	mean $\pm$ SE
HumanEval	+6.9	+6.6	+3.3	+5.6 $\pm$ 1.2
GSM8K	+7.1	+8.4	+1.8	+5.7 $\pm$ 2.0
MT-Bench	+3.5	−0.2	+2.7	+2.0 $\pm$ 1.1

gains +6.4%, +7.5% and +2.9% on Llama-3.1-8B at adaptive mean draft 2.3. On Qwen3-14B it is a tie to slight win, +1.3%, +0.3% and +0.7% at mean draft 3.05. The gentler cum(0.05), which drafts to a mean of 4.22 there, loses instead: −4.3%, −7.3% and −4.9%.

Fair baselines. Padding the fixed arms to 7 slots could in principle hand the adaptive arm a verification advantage it has not earned, which is failure mode 6. So we loaded three engines in one process, native $K=2$ and $K=3$ each with its true verification budget plus the patched $K=7$ cum engine, and round-robin across engines on three fresh containers under vLLM v0.24. Each native engine provably drafts exactly its K , and realized draft lengths are bit-identical across containers, since greedy decoding is deterministic and only the clock varies. The result is positive or tied in 9 of 9 workload cells (Table VI). Cross-container spread exceeds the within-run error bars, with MT-Bench alone spanning −0.2 to +3.5 across three identically configured containers. That is the strongest argument in this paper against trusting any single-container benchmark number, including ours.

The padded and native baselines agree because verification is flat in K , and we measured that rather than assuming it. A verify microbench under HF-eager isolation shows one kernel-regime step, from the non-speculative single-token forward at $q=1$, 15.8 ms, to the multi-token path. After that it is flat: $q=2$ to $q=8$ runs 18.5 down to 18.2 ms at $B=1$. Native $K=2$ ’s $q=3$ and the padded arm’s $q=8$ differ by under 1%. The honest headline for Table VI is therefore +2 to 6% with one of three workloads a wash: MT-Bench reads +2.0 $\pm$ 1.1 across three containers, which three containers cannot separate from zero, its worst cell reads −0.2 $\pm$ 2.0 within its own run, and no cell is a significant loss.

Temperature and batch. At $T=0.8$ with seeded sampling and rejection-sampling verification the win survives, +5.4%, +2.1% and +1.5%, all positive-significant, at realized depth 2.24. Sampling lowers acceptance, so saturation fires earlier, which is the direction the mechanism predicts. With batch the gain decays monotonically (Fig. 4): +2 to 6% at $B=1$, about +1% at $B=4$, and 0 to −3% at $B=8$. Realized depth moves 2.30, 2.68, 2.57 across the three batch sizes, rising where the gain first erodes rather than tracking it all the way down. We use a batch-level mean-confidence rule because per-request ragged stopping is impossible in a batch-synchronized draft

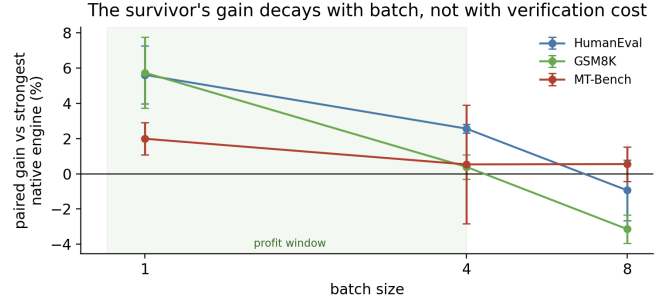


Fig. 4. The survivor’s profit window. Paired gain over the strongest native engine decays from +2 to 6% at $B=1$ to between 0 and −3% at $B=8$. The verify microbench rules out verification-cost growth as the cause; the batch-synchronized mean stop is what erodes it.

loop, which is itself a finding about adaptive-length methods in serving engines. The microbench identifies the mechanism: verify latency stays flat even at $B=8 \times q=8$, so the decay is not verification-cost growth. It is the batch-synchronized mean stop wasting per-request adaptivity, forcing rejects on requests whose chains were still alive, and the rising realized depth is that compromise made visible.

The practitioner’s question is whether this can be switched on in production, and for a batched deployment our answer is no, with one qualification.

The qualification is that we tested the batch-synchronized version, not a ragged engine that stops each request independently. Such an engine is buildable and might decay differently. We did not build one, so the deployable number is open rather than settled. What we can claim is that the profit window we observed is latency-sensitive low-batch serving, $B \lesssim 4$, consistent with our own batch-collapse result (§IX) and with SpecDecode-Bench’s batch-dependent optimal draft lengths [18]. Both directions were stated in advance of the runs as consequences of the saturation mechanism.

None of this contradicts the methods that do pay off in production. Nightjar [16] and TurboSpec [17] adapt at the load level, deciding whether to speculate at all given batch size and system load, which is a different lever from per-step draft length. Our batch-collapse data supports that choice: by $B \geq 32$ the best- K gap is exactly zero, so there is nothing left for a per-step controller to win, while the decision of whether to speculate still has room in it.

Pulling the settings together (Fig. 5): a free, signal-less saturation-pruning policy at its train-selected τ ties or beats the tuned fixed baseline everywhere we tested. Llama chain +2.9 to 7.5%, Llama tree +2 to 5%, Qwen3 chain +0.3 to 1.3%, weak head about zero. Every learned acceptance-prediction policy loses everywhere.

Adaptive *deepening* beyond the default budget is a different lever and we did not test it (cf. TALON [15]).

IX. SUPPORTING RESULTS

Batch collapse. Adaptive headroom shrinks as batch grows. At $B=16$ the best- K gap over $K=1$ is still +5 to 11%; at

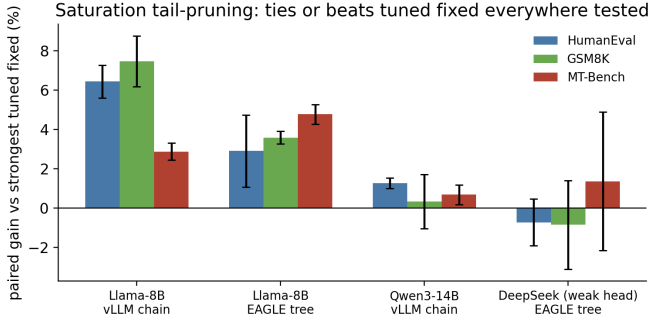


Fig. 5. The survivor across settings. Saturation tail-pruning ties or beats the strongest tuned fixed baseline everywhere tested, and the gain scales with draft-head strength.

TABLE VII

ONLINE *per-request* CONTROLLERS AGAINST A TUNED FIXED DEPTH: EACH POLICY PICKS ONE K PER PROMPT AND ADAPTS ACROSS PROMPTS, WHICH IS A DIFFERENT LEVER FROM THE PER-STEP STOPPING POLICIES OF §VII. RUN IN A SEPARATE HARNESS AND WORTH DISTINGUISHING FROM THE REST OF THE WALL-CLOCK WORK: VLLM 0.23 IN THE CHAIN REGIME WITH `NUM_SPECULATIVE_TOKENS=7`, RATHER THAN THE SAFEAILAB EAGER TREE IMPLEMENTATION, AND OVER 90 PROMPTS (30 PER WORKLOAD ACROSS HUMAN-EVAL, GSM8K AND MT-BENCH) RATHER THAN THE 10 HELD-OUT PROMPTS THE §VII TABLES USE. LLAMA-3.1-8B WITH EAGLE-3, H100, $B=1$, GREEDY, MEASURED WALL-CLOCK. EVERY REACTIVE CONTROLLER LOSES, AND THE PER-REQUEST ORACLE BOUNDS WHAT ANY OF THEM COULD HAVE WON AT +2.0%.

Policy	net speedup	vs best fixed
Fixed $K=2$ (tuned baseline)	1.538×	—
UCB (BanditSpec-style)	1.463×	−4.9%
ϵ -greedy	1.432×	−6.9%
Acceptance-history	1.394×	−9.3%
Oracle (per-request)	1.569×	+2.0%

$B \geq 32$ it is exactly zero on every workload. The serving regime removes even the theoretical motivation for per-step control. This also bounds the gate-3 result, since verification dominance grows with batch, so tail-pruning’s draft-side savings can only shrink. The +2 to 6% is a $B=1$ ceiling.

Long context. Headroom at 8K is gated by draft quality, not KV bandwidth. Qwen3-14B holds 1.31 to 1.39× at $\text{ctx}=8192$, and even at $B=32$ keeps a +3% gap, the one cell consistent with MagicDec [19]. Llama-8B and DeepSeek collapse to about 1.0× as their heads’ acceptance falls to roughly 1.1 tokens per step.

Reconciling prior gains. SVIP [6], AdaEDL [10], SpecDec++ [7] and DDD [12] report 2 to 8% against default- K or $K=1$ baselines. Against our tuned fixed K the per-request oracle, which is the relevant bound for a method that effectively selects one depth per request, is only +2.0%, and every online controller we ran went negative (Table VII).

We report these in some detail because the failure has a cause worth sharing, and because our implementations are simple enough that someone with a better one may well do otherwise. Two measurements from this setting explain the

pattern. First, the prize is capped: the per-request oracle is +2.0%, so a controller has to find the right depth almost immediately or its exploration is unrecoverable. Reactive bandits explore by construction, and here there is nothing to amortise that exploration against. Second, the quantity they are trying to learn barely persists. The signal capture behind this harness covers 1,295 decode steps. Across them the accept-run distribution is heavily zero-inflated, with 61% of steps accepting no draft token at all, and the lag-1 autocorrelation of the accept-run is only 0.247. What the last step taught is a weak guide to the next one, so an online learner spends its budget tracking something close to noise.

That is the learning we would most like others to test. A stronger bandit, a better-designed exploration schedule, or a setting with higher autocorrelation might change this. We would be glad to see it, and the traces are released so that a counter-example can be produced against the same data rather than a new capture.

We read this as a baseline effect, not an error in those papers. Their measurements are correct for the baselines they used, and moving off a poorly chosen fixed depth is a real improvement worth reporting. The difficulty is that tuning the fixed depth captures much of the same improvement, so the two do not add: a practitioner who tunes K first should not expect the published increment on top. What our decomposition adds is an account of why the remainder is so thin once tuning is done, and that account applies to our own probe as much as to anyone else’s method.

X. DISCUSSION

Our advice to anyone setting out to shorten drafts from a learned signal is to compute the ceiling before building the predictor. The gap between a tuned fixed depth and the per-step oracle is the number that motivates the field, and it is the wrong number to aim at. Split it first into the part a draft-side signal could reach and the part that is verification-outcome variance no draft-side signal can see. On every model pair we measured, the second part is about four fifths of the total, and what remains is small enough that a predictor has to be nearly free to be worth running. That split costs one capture run and an afternoon of analysis. It is much cheaper than the months we spent finding out the other way, and on our pairs it would have said no.

Two things this advice does not cover. It concerns the shortening lever only: adaptive deepening beyond the default budget is untested here (cf. TALON [15]). And it concerns draft-side signals only, since a quantity available after verification is a different question, if not a usable one for this purpose.

The oracle plots themselves are real. They are simply a measure of hindsight, and the reachable slice inside them is genuinely present in the hidden state, worth a few percent offline and negative end-to-end once a learned policy has to pay for itself. The one adaptive lever that pays needs no signal at all.

That pattern explains something about where the field went in 2025 and 2026. Work moved toward load-level adaptation,

deciding whether to speculate based on batch and system load, and our batch-collapse data confirms that is the profitable lever. Signal-level length controllers keep failing against tuned baselines for a structural reason, not because nobody has found the right feature yet.

For practitioners, the advice is short. Ship a tuned, deep fixed draft length plus free saturation tail-pruning, and spend the adaptivity budget at the serving level instead of on per-step prediction. For anyone evaluating such methods, at the $\pm 5\%$ scale these claims occupy, mechanism-matched simulators, held-out threshold selection and paired drift-controlled timing are not optional extras.

XI. LIMITATIONS

- Offline gains are cost-model figures at $B=1$: net $+2.0$ to 3.1% , recovery $+16.8 \pm 2.7\%$ and $+17.2 \pm 2.4\%$ for the two instruct models under the corrected paired estimator. End-to-end, learned early-stopping does not realize them.
- The probe is trained in-distribution, on held-out generations from the same models and workloads. Cross-workload and cross-model transfer is untested.
- The reasoning-head null is confounded by lower base acceptance (§V) and is scoped to that head.
- “Irreducible” means irreducible from draft-side signals. Privileged post-verification quantities are a different question.
- Gate 3 tests early stopping only. Adaptive deepening beyond the default budget is untested. Single-pass timings in §VII’s first two experiments carry a noise bar of about 2% ; the policy-zoo verdict uses the paired protocol.
- The tail-pruning positive holds for a strong instruction head on an H100 at $B=1$, with HumanEval borderline, and ties on the weaker DeepSeek-R1-Distill head. The gain is head-dependent and scoped accordingly.
- Wall-clock work uses the eager reference implementation, since vLLM has no per-step hook. Relative fixed-versus-adaptive comparisons are valid; absolute tok/s is lower than vLLM. This also bounds the probe-overhead measurement in §VIII: a per-level Python call is near its worst case in eager mode, and a fused or batched implementation would cost less. The direction of that result is safe, since the probe must do strictly more work than reading a number the tree already holds, but the 10 to 13% magnitude is specific to this engine.
- The wall-clock samples are small: 10 held-out prompts per workload at 128 max new tokens, 8 in the batched runs. Precision comes from repetition and pairing rather than from prompt count, and the reported error bars are over cycles, not over prompts. Cross-container spread can exceed the within-run bars: MT-Bench reads $+3.5 \pm 0.3$ on one container and -0.2 ± 2.0 on another, a gap wider than either run could see from inside itself. That is the honest statement of how far these magnitudes should be trusted.

XII. CONCLUSION

We followed the speedup that adaptive draft length promises and traced where it is lost. Against a tuned fixed draft length, about 80% of the celebrated per-step oracle is irreducible from any draft-side signal. The reducible fifth lives in the drafter’s hidden state, is worth $+2$ to 3% offline for instruction-model heads, and loses 2 to 17% end-to-end when a policy acts on it, because verification dominates execution. We audited three published-style stopping policies on the clock, and re-ran the one that survived under a paired, drift-controlled protocol that six evaluation failure modes forced on us. Exactly one policy is left standing: signal-free saturation tail-pruning, worth $+2$ to 6% at zero model cost, with one of three workloads a wash and only at low batch. The deployable recommendation is a tuned deep draft plus free tail-pruning. The learned per-step signal the field keeps chasing is real, we know where it lives, and for the shortening lever it is not worth acting on.

REPRODUCIBILITY

Everything is released at <https://github.com/pramodjella/speculative-decoding-draft-length-audit>, including the raw per-run JSON behind every table and figure here except Table III, whose harness writes to a Modal volume; its cells are recorded in `DELIVERABLES.md`, which maps each result (E1–E6) to the script and artifact that produced it.

- Hidden-state capture: `modal_eagle3_hidden_full_capture.py`
- Decomposition: `analyze_bayes_ceiling_paired.py`, the paired-within-fold version carrying the nesting and position self-tests, with its permutation control and independent re-derivation
- Cost grounding: `measured_cost_ground.py`
- Wall-clock harnesses: `modal_eagle_wallclock.py`, `modal_eagle_inloop.py`
- Policy zoo and paired protocol: `modal_eagle_zoo_verify.py`, `modal_eagle_paired.py`
- Fair-baseline native-engine replication: `modal_vllm_fairbase.py`
- Per-request controllers (Table VII): `modal_controller_benchmark.py`

A single `scripts/check_stale.sh` gate guards the shipped documents against retired numbers.

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